

SkyLink Airways — UML Class Diagram

Class / Relationship	Main members / methods
Flight (abstract)	flightNumber, origin, destination, departureDateTime, totalSeats, availableSeats; calculateAvailableSeats()
DomesticFlight	routeFactor; overrides fare/display/serialization
InternationalFlight	visaRequired; overrides fare/display/serialization
CharterFlight	contractHolder; overrides fare/display/serialization
Passenger (abstract)	passengerId, name, phone; baggageAllowance(), loyaltyMultiplier(), cancellationPercentage()
EconomyPassenger	20 kg baggage; 1.0x loyalty; 60% base cancellation refund
BusinessPassenger	30 kg baggage; 1.5x loyalty; 80% base cancellation refund
FirstClassPassenger	40 kg baggage; 2.0x loyalty; 90% base cancellation refund
Ticket	ticketId, passengerId, flightNumber, seatNumber, farePaid, status; cancel(); == and <<
Airline	owns flights/passengers; stores tickets; booking, cancellation, reports, persistence

Inheritance: Flight → DomesticFlight / InternationalFlight / CharterFlight; Passenger → EconomyPassenger / BusinessPassenger / FirstClassPassenger. Airline aggregates Flight, Passenger and Ticket. Ticket references a passenger and flight by identifier.